Simbarashe Aldrin Ngorima

Curriculum Vitae

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RESEARCH PROFILE

Postdoctoral researcher at the MUST Deep Learning research group, Faculty of Engineering, North-West University, affiliated with the Centre for Artificial Intelligence Research (CAIR) and with NITheCS, where I held a doctoral scholarship from 2022 to 2025. My research develops gradient-based attribution methods and structured compression techniques for deep neural networks applied to time-series problems. The PhD focused on IEEE 802.11p vehicular channel estimation, resulting in five peer-reviewed publications and three IEEE journal submissions currently under review. The postdoctoral work extends the same methodology to deep learning for prosthetics, including gait phase prediction and metabolic cost estimation, and to inter-subject domain adaptation for wearable sensing.

EDUCATION

Ph.D., Computer and Electronic Engineering. North-West University, South Africa. *2022 to 2025.*
Thesis: "Channel Estimation for IEEE 802.11p Vehicular Communication using Deep Learning." Supervisors: Prof. Marelie H. Davel and Prof. Albert S.J. Helberg. Funded by the NITheCS Doctoral Scholarship, the Telkom Centre of Excellence, and CAIR. Result: passed.

M.Eng., Computer and Electronic Engineering. North-West University, South Africa. *2020 to 2021.*
Funded by the National Research Foundation Master's Scholarship.

B.Eng., Electronic Engineering. Parul University, India. *2015 to 2019.*

ACADEMIC APPOINTMENTS AND AFFILIATIONS

Postdoctoral Researcher. MUST Deep Learning, Faculty of Engineering, North-West University. *2026 to present.*

- Postdoctoral research on deep learning for prosthetics, with primary focus on gait phase prediction and metabolic cost estimation. Affiliated with the Centre for Artificial Intelligence Research (CAIR) and with NITheCS.
- Continuing collaborative research with my doctoral supervisors on interpretability-driven compression of deep learning models, with three IEEE journal submissions currently under review.

Doctoral Researcher. MUST Deep Learning, Faculty of Engineering, North-West University. *2022 to 2026.*

- Doctoral research on deep learning for IEEE 802.11p vehicular channel estimation, supervised by Prof. Marelie H. Davel and Prof. Albert S.J. Helberg. Funded by NITheCS, the Telkom Centre of Excellence, and CAIR.

RESEARCH INTERESTS

- Interpretability of deep neural networks
- Structured neural network compression: filter pruning, attribution-guided sparsification, and near-optimal weight quantisation.
- Deep learning for wireless communications and applied signal processing.
- Cross-domain transfer of interpretability and compression methods to biomedical time-series, including gait analysis and wearable sensing.
- On-device inference for V2X, Internet of Things, and resource-constrained edge platforms.

PUBLICATIONS

Journal and conference articles under review

1. Ngorima, S.A., Helberg, A., and Davel, M.H. "Convolutional DPA Refinement for Real-Time Channel Estimation in High-Mobility Vehicular Networks" *PhD core contribution, submitted to IEEE Access*.
2. Ngorima, S.A., Helberg, A., and Davel, M.H. "REACH: Interpretability-Driven Feature Identification and Architecture Compression for Multi-Channel Vehicular Channel Estimation." IEEE Transactions on Machine Learning in Communications and Networking. *Submitted 2026*. Available as arXiv preprint arXiv:2606.11857.
3. Ngorima, S.A., Helberg, A., and Davel, M.H. "Joint Interpretability-Guided Pruning and Near-Optimal Weight Quantisation for Deep Vehicular Channel Estimators." IEEE GLOBECOM 2026. *Under review*.

Refereed conference and book-chapter publications

4. Ngorima, S.A., Helberg, A., and Davel, M.H. "Simplified Temporal Convolutional Based Channel Estimation for a WiFi Vehicular Communication Channel." IEEE VTS Wireless Africa Conference, 2025.
5. Ngorima, S.A., Helberg, A., and Davel, M.H. "Feature Extraction for Plant Growth Estimation." Communications in Computer and Information Science, vol. 2784, Springer, Cham, 2025.
6. Ngorima, S.A., Helberg, A., and Davel, M.H. "Neural Network-Based Vehicular Channel Estimation Performance: Effect of Noise in the Training Set." Communications in Computer and Information Science, vol. 2326, Springer, Cham, 2024.
7. Ngorima, S.A., Helberg, A., and Davel, M.H. "A Data Pilot-Aided Temporal Convolutional Network for Channel Estimation in IEEE 802.11p Vehicle-to-Vehicle Communications." Southern Africa Telecommunication Networks and Applications Conference (SATNAC), 2024.
8. Ngorima, S.A., Helberg, A., and Davel, M.H. "Sequence-Based Deep Neural Networks for Channel Estimation in Vehicular Communication Systems." Communications in Computer and Information Science, vol. 1976, Springer, Cham, 2023.

CONFERENCE PRESENTATIONS

- IEEE VTS Wireless Africa Conference. Presentation in 2025.
- Southern African Conference for Artificial Intelligence Research (SACAIR). Presentations in 2023, 2024 and 2024.
- Southern Africa Telecommunication Networks and Applications Conference (SATNAC), 2024.

RESEARCH FUNDING AND AWARDS

- Award for publishing five publications during PhD studies, North-West University, 2026.
- NITheCS Doctoral Scholarship, 2022 to 2025.
- Telkom Centre of Excellence research support, 2022 to 2025.
- Centre for Artificial Intelligence Research (CAIR) support, 2022 to 2025.
- National Research Foundation Master's Scholarship, 2021.

TEACHING AND MENTORSHIP

- Teaching 3rd year module (Signal Theory) at North-West University - second semester 2026.
- Currently supervising final year (4th year) Computer and Electronic Engineering students at North-West University.
- Available to contribute to: deep learning for time-series, machine learning for communications, signal processing, applied probability and statistics, scientific programming in Python and MATLAB, and capstone or final-year design projects.

ACADEMIC AND PROFESSIONAL SERVICE

- Communications Chairperson, Deep Learning Indaba, 2023 to present. Responsible for communications channels and content production for the largest African machine learning organisation.
- Communications Champion, MUST Deep Learning initiative, 2023 to present.
- Member, MUST Deep Learning Group, North-West University, 2022 to present.

TECHNICAL SKILLS

- Deep learning frameworks: PyTorch (primary), TensorFlow, scikit-learn, NumPy, pandas, h5py.
- Programming: Python, MATLAB, C/C++.
- Methods: temporal convolutional networks, sequence modelling, structured pruning, near-optimal weight quantisation (Lloyd-Max), gradient-based attribution, hyperparameter optimisation (Optuna), and domain adaptation.
- Wireless and DSP: OFDM, IEEE 802.11p, V2V and V2I channel modelling, BER and NMSE analysis.
- Tools and workflow: Git, LaTeX, VS Code, PyCharm, Jupyter, Linux.
- Spoken languages: English (Fluent), Shona (Native).

REFERENCES

Prof. Marelie H. Davel
Research Professor, Director - MUST Deep Learning, Faculty of Engineering, North-West University.
South African Research Chair (SARChI) in Deep Learning.
Email: marelie.davel@nwu.ac.za
Relationship: Doctoral supervisor and current postdoctoral collaborator.

Prof. Albert S.J. Helberg
Professor, School of Electrical, Electronic and Computer Engineering, North-West University.
Research Leader, TeleNet Research Group, Telkom Centre of Excellence.
Email: albert.helberg@nwu.ac.za
Relationship: Doctoral co-supervisor and current postdoctoral collaborator.